Network Reconnaissance & Port Scanning Audit

Ethical reconnaissance and vulnerability scanning with Nmap and Wireshark

RabTech Academy | Cybersecurity & Ethical Hacking | Task 03

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Target (simulated)	127.0.0.2 - isolated loopback sandbox host (lab.local)
Scanner host	127.0.0.1 (same machine, isolated)
Tools	Nmap 7.94SVN, tcpdump, TShark/Wireshark 4.2.2
Classification	Training exercise - no production systems touched

1. Executive Summary

This audit performed structured reconnaissance against an authorized, simulated sandbox host (127.0.0.2) built specifically for this exercise. A full-range TCP SYN scan, a UDP scan of the top 100 ports, service/version detection, OS detection and default NSE scripts were run with Nmap while all traffic was recorded for analysis in Wireshark.

Key results: 7 TCP ports and 2 UDP ports were found open on the target (SSH, FTP, Telnet, SMTP, HTTP, MySQL, plus DNS and SNMP over UDP). Packet analysis recovered **three sets of valid credentials in plain text** - over FTP, Telnet and HTTP Basic authentication - demonstrating that anyone able to observe the network path could take over these accounts. The database port is reachable from the network and several services disclose exact product versions.

Severity	Count	Findings
High	4	Cleartext Telnet, cleartext FTP, HTTP Basic auth without TLS, SNMP community 'public'
Medium	3	Exposed MySQL, outdated/EOL software versions, excessive attack surface
Low / Info	3	Version banners, SMTP host disclosure, DNS over UDP responding

2. Scope, Authorization and Rules of Engagement

Item	Detail
In scope	Single host 127.0.0.2 (sandbox target created for this exercise). All TCP ports 1-65535; UDP top 100 ports.
Out of scope	Every other host, network or internet system. No exploitation, brute forcing, denial of service or data modification.
Authorization	Self-owned lab. The target services were created and started by the tester inside an isolated environment.
Owner / contact	Tester (Vignesh M) - acts as both system owner and auditor in this training scenario.
Risk controls	Loopback-only traffic, no external packets; non-intrusive scan timing (-T4); only the lab's own test accounts were used to generate authentication traffic.
Stop condition	Stop immediately if any non-lab address responds or any service becomes unstable.

3. Methodology and Lab Environment

The workflow followed standard phases: (1) lab preparation, (2) host and port discovery, (3) service and OS fingerprinting, (4) passive packet analysis, (5) risk analysis and remediation. The sandbox target emulates a small Linux server with deliberately weak services: vsFTPd-style FTP (21), OpenSSH-style SSH (22), Telnet login (23), Postfix-style SMTP (25), an Apache-style web admin panel with HTTP Basic auth (80), a MySQL-style listener (3306), and UDP DNS (53) and SNMP (161) responders. A packet capture (tcpdump on the loopback interface) ran during every scan and was later opened in Wireshark/TShark.

Commands executed

```
nmap -sS -p- -T4 127.0.0.2 -oN syn.txt                    # TCP SYN, all 65535 ports
nmap -sU --top-ports 100 -T4 127.0.0.2 -oN udp.txt        # UDP scan
nmap -sV -p 21,22,23,25,53,80,3306 127.0.0.2 -oN ver.txt  # service versioning
nmap -O --osscan-guess -p 21,22,80 127.0.0.2 -oN os.txt   # OS detection
nmap -sC -p 21,23,80 127.0.0.2 -oN scripts.txt            # default NSE scripts
tcpdump -i lo -w scan.pcap host 127.0.0.2                 # capture during scans
```

4. Nmap Scan Results

4.1 TCP SYN scan (all ports)

```
Nmap scan report for 127.0.0.2
Host is up (0.0000030s latency).
Not shown: 65527 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
80/tcp   open  http
2024/tcp open  xinuexpansion4
2025/tcp open  ellpack
3306/tcp open  mysql
```

Eight TCP ports answered. Ports 2024 and 2025 belong to the lab container's own runtime and are not part of the target design; they are excluded from the findings. The remaining seven are the audited services.

4.2 UDP scan (top 100 ports)

```
Nmap scan report for 127.0.0.2
Host is up (0.0000040s latency).
Not shown: 98 closed udp ports (port-unreach)
PORT    STATE SERVICE
53/udp  open  domain
161/udp open  snmp
```

4.3 Service version detection

```
Nmap scan report for 127.0.0.2
Host is up (0.000022s latency).

PORT     STATE  SERVICE VERSION
21/tcp   open   ftp
22/tcp   open   ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
23/tcp   open   telnet?
25/tcp   open   smtp    Postfix smtpd
53/tcp   closed domain
80/tcp   open   http    Apache httpd 2.4.41 ((Ubuntu) Python/3.12.3)
3306/tcp open   mysql   MySQL 5.7.42-0ubuntu0.18.04.1
```

Nmap could not match the FTP and Telnet fingerprints to its database (it prints long fingerprint submissions, omitted here); their banners were read manually from the packet capture instead (vsFTPd 3.0.3 banner on FTP, Ubuntu 20.04.6 LTS login banner on Telnet). Nmap also inferred *Service Info: Host mail.lab.local; OS: Linux*.

4.4 OS detection

```
Nmap scan report for 127.0.0.2
Host is up (0.000066s latency).

PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (97%), Linux 3.1 (95%), Linux 3.2 (95%), AXIS 210A or 211 Network Camera (Linux 2.6.
No exact OS matches for host (test conditions non-ideal).
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

Interpretation: Nmap's guesses point to the Linux kernel family, consistent with the Ubuntu banners. Confidence is low because the scan ran over loopback (0 network hops, no closed port in the tested set), which Nmap warns about. On a real network this step would be repeated with both open and closed ports.

4.5 Default NSE scripts

```
Nmap scan report for 127.0.0.2
Host is up (0.000039s latency).

PORT   STATE SERVICE
21/tcp open  ftp
| ftp-syst:
|   STAT:
| FTP server status:
|  Connected to: 127.0.0.2:21
|  Waiting for username.
|  TYPE: ASCII; STRUcture: File; MODE: Stream
|  Data connection closed.
|_End of status.
23/tcp open  telnet
80/tcp open  http
|_http-title: Site doesn't have a title.
| http-auth:
| HTTP/1.0 401 Unauthorized\x0D
|_  Basic realm=Admin Panel
```

5. Wireshark Packet Analysis

Captures were opened in Wireshark/TShark (scan.pcap for the scans, creds.pcap for authentication traffic). The figures below are the packet list, stream and detail views of those captures.

5.1 How the scans look on the wire

Wireshark view - scan.pcap (Nmap -sS, attacker 127.0.0.1 -> target 127.0.0.2)

Apply a display filter: `tcp  (first 14 frames of capture)`

No.	Time	Source	Destination	Proto	Len	Info
1	0.000000	127.0.0.1	127.0.0.2	TCP	58	61739 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2	0.000011	127.0.0.2	127.0.0.1	TCP	54	443 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
3	0.000016	127.0.0.1	127.0.0.2	TCP	58	61739 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
4	0.000018	127.0.0.2	127.0.0.1	TCP	54	3389 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
5	0.000021	127.0.0.1	127.0.0.2	TCP	58	61739 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6	0.000023	127.0.0.2	127.0.0.1	TCP	54	1723 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	0.000026	127.0.0.1	127.0.0.2	TCP	58	61739 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
8	0.000028	127.0.0.2	127.0.0.1	TCP	54	5900 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
9	0.000031	127.0.0.1	127.0.0.2	TCP	58	61739 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
10	0.000033	127.0.0.2	127.0.0.1	TCP	54	587 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
11	0.000036	127.0.0.1	127.0.0.2	TCP	58	61739 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
12	0.000038	127.0.0.2	127.0.0.1	TCP	54	139 → 61739 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
13	0.000041	127.0.0.1	127.0.0.2	TCP	58	61739 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14	0.000052	127.0.0.2	127.0.0.1	TCP	58	23 → 61739 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495

Closed ports answer SYN with RST/ACK (red rows). Frame 14: port 23 answers SYN/ACK (highlighted) = OPEN.

Fixed source port 61739 and Win=1024 on every probe are the classic Nmap half-open (SYN) scan signature.

Figure 1. TCP SYN scan. Each probe is a bare SYN; closed ports return RST/ACK, open ports return SYN/ACK. The scanner never completes the handshake (half-open), which is why SYN scans are quieter than connect scans but still easy to spot in a capture.

Across the capture, the full-range scan generated one SYN probe per port (65,535) and received SYN/ACK only from ports 21, 22, 23, 25, 80, 3306 and the two runtime ports - matching Nmap's report exactly. A sudden burst of thousands of SYNs from one source with sequential or random destination ports and near-zero payload is the pattern an IDS should alert on.

Wireshark view - scan.pcap (Nmap -sU --top-ports 100)

Apply a display filter: `(udp.port == 53 || udp.port == 161 || icmp.type == 3)  - representative frames`

No.	Time	Source	Destination	Proto	Len	Info
131080	0.771999	127.0.0.2	127.0.0.1	ICMP	70	Destination unreachable (Port unreachable) - port CLOSED
131082	0.772012	127.0.0.2	127.0.0.1	ICMP	82	Destination unreachable (Port unreachable) - port CLOSED
131084	0.772019	127.0.0.2	127.0.0.1	ICMP	142	Destination unreachable (Port unreachable) - port CLOSED
131301	0.772799	127.0.0.1	127.0.0.2	UDP	72	43801 -> 53 Len=30 (probe)
131302	0.772875	127.0.0.2	127.0.0.1	UDP	72	53 -> 43801 Len=30 (RESPONSE: port OPEN)
131307	0.772916	127.0.0.1	127.0.0.2	UDP	93	43801 -> 161 Len=51 (probe)
131308	0.772930	127.0.0.2	127.0.0.1	UDP	70	161 -> 43801 Len=28 (RESPONSE: port OPEN)

Replies from UDP/53 and UDP/161 mark those ports OPEN; ICMP 'Port unreachable' marks closed UDP ports.

Figure 2. UDP scan. A reply from the service proves the port is open; an ICMP port-unreachable proves it is closed. UDP ports that stay silent are reported as open|filtered, which is why UDP results need cross-checking.

5.2 Unencrypted credentials

Wireshark view - creds.pcap : FTP session

Apply a display filter: `ftp`

No.	Time	Source	Destination	Proto	Len	Info
4	0.000421	127.0.0.2	127.0.0.1	FTP	90	Response: 220 vsFTPd 3.0.3 ready
6	0.000559	127.0.0.1	127.0.0.2	FTP	80	Request: USER labuser
8	0.000619	127.0.0.2	127.0.0.1	FTP	99	Response: 331 Username ok, send password.
9	0.000639	127.0.0.1	127.0.0.2	FTP	83	Request: PASS labpass123
10	0.000697	127.0.0.2	127.0.0.1	FTP	89	Response: 230 Login successful.
11	0.000713	127.0.0.1	127.0.0.2	FTP	74	Request: TYPE A
12	0.000809	127.0.0.2	127.0.0.1	FTP	91	Response: 200 Type set to: ASCII.
13	0.000829	127.0.0.1	127.0.0.2	FTP	72	Request: PASV
14	0.000984	127.0.0.2	127.0.0.1	FTP	114	Response: 227 Entering passive mode (127,0,0,2,187,195).
18	0.001360	127.0.0.1	127.0.0.2	FTP	72	Request: LIST

Frames 6 and 9: USER and PASS travel in plaintext. Username 'labuser' / password 'labpass123' recovered.

Figure 3. FTP login captured with the filter `ftp`. The USER and PASS commands are readable in the Info column.

```
Wireshark view - creds.pcap : Telnet (tcp.port == 23)

Follow TCP Stream (tcp.stream) - 127.0.0.1:34330 <-> 127.0.0.2:23   [red = client, blue = server]


Ubuntu 20.04.6 LTS
login:
admin
Password:
Winter2026!

Last login: Mon Oct  5 10:11:12
$

Result: Telnet username 'admin' and password 'Winter2026!' captured in clear text.
```

Figure 4. 'Follow TCP Stream' on the Telnet session. Both the login name and password appear exactly as typed.

```
Wireshark view - creds.pcap : packet details pane, HTTP Basic authentication

Frame 53: 235 bytes on wire  |  127.0.0.1 -> 127.0.0.2  |  Transmission Control Protocol, Dst Port: 80
Hypertext Transfer Protocol
  GET /admin HTTP/1.1\r\n
  Host: 127.0.0.2\r\n
  Authorization: Basic d2ViYWRtaW46UGFzc3cwcmQjMQ==\r\n
      Credentials: webadmin:Passw0rd#1        <-- decoded by Wireshark (Base64 is encoding, not encryption)
  [Full request URI: http://127.0.0.2/admin]

Frame 59: HTTP/1.0 200 OK (text/html)   Server: Apache/2.4.41 (Ubuntu)
  -> Server banner discloses product and exact version.
```

Figure 5. HTTP Basic authentication. The header carries Base64 text that Wireshark decodes instantly; Base64 is an encoding, not encryption.

Protocol	Port	Wireshark filter used	Credential recovered (lab test account)
FTP	21/tcp	ftp.request.command == "USER" || ftp.request.command == "PASS"	labuser / labpass123
Telnet	23/tcp	tcp.port == 23 (Follow TCP Stream)	admin / Winter2026!
HTTP Basic	80/tcp	http.authorization	webadmin / Passw0rd#1

These are throw-away accounts created for the lab; they are shown to document the finding.

6. Open Ports, Services and Potential Attack Vectors

Port	Service / version	Observed	Potential attack vectors
21/tcp	FTP - vsFTPd 3.0.3 banner	Plaintext USER/PASS	Credential sniffing, brute force, anonymous-access checks, file tampering if write enabled
22/tcp	SSH - OpenSSH 8.2p1 Ubuntu	Banner discloses version	Password brute force, user enumeration, exploitation if version is unpatched
23/tcp	Telnet - Ubuntu 20.04.6 LTS login	Plaintext login and session	Sniffing, session hijacking, man-in-the-middle, default credential guessing
25/tcp	SMTP - Postfix, host mail.lab.local	Banner reveals software and hostname	User enumeration (VRFY/RCPT), open-relay testing, spoofing / phishing relay
80/tcp	HTTP - Apache 2.4.41 (Ubuntu)	Basic auth over HTTP, 401 realm 'Admin Panel'	Credential interception, admin-panel brute force, outdated-server exploits
3306/tcp	MySQL 5.7.42	Database reachable from network	Remote login brute force, privilege abuse, exploitation of an end-of-life version
53/udp	DNS	Responds to queries	Open-resolver abuse / amplification if recursion is on, information leakage
161/udp	SNMP	Answers with community 'public'	Device and configuration enumeration; write access if a writable community exists

7. Findings and Risk Rating

ID	Finding	Severity	Evidence
F-01	Telnet enabled; credentials sent in clear text	High	Fig. 4, Nmap 23/tcp open
F-02	FTP without encryption; credentials in clear text	High	Fig. 3, Nmap 21/tcp open
F-03	Admin panel uses HTTP Basic auth without TLS	High	Fig. 5, Nmap http-auth script
F-04	SNMP responds to default community 'public'	High	Nmap UDP scan, Fig. 2
F-05	MySQL exposed to the network; version 5.7 is past end of life	Medium	Nmap -sV, 3306/tcp
F-06	Outdated service versions (e.g. Apache 2.4.41) - check against NVD for known CVEs	Medium	Nmap -sV
F-07	Excessive attack surface: 9 reachable services on one host	Medium	SYN + UDP scans
F-08	Version and hostname disclosure in banners	Low	Nmap -sV, mail.lab.local
F-09	DNS (UDP) answering - confirm recursion is restricted	Low	Nmap UDP scan
F-10	OS family identifiable (Linux)	Info	Nmap -O

Severity reflects likelihood and impact assuming an attacker with network visibility; it was assigned by the auditor and is not a CVSS calculation.

8. Recommendations

Priority	Action
Immediate	Disable Telnet and FTP; use SSH with key authentication and SFTP instead (F-01, F-02).
Immediate	Serve the admin panel only over HTTPS (TLS 1.2+), add a stronger login than Basic auth, restrict by IP/VPN (F-03).
Immediate	Disable SNMP, or move to SNMPv3 with authentication and encryption; change the default community and limit sources (F-04).
Short term	Bind MySQL to localhost or a private management network and firewall port 3306 (F-05).
Short term	Patch or upgrade Apache, OpenSSH and MySQL to supported releases; subscribe to vendor advisories (F-06).
Short term	Apply a default-deny host firewall exposing only required ports; remove unused services such as SMTP if not needed (F-07).
Hardening	Hide version strings (ServerTokens Prod, Postfix smtpd_banner, SSH banner) and restrict DNS recursion (F-08, F-09).
Monitoring	Alert on SYN-scan patterns and on any cleartext-credential protocol seen on the network; re-scan after fixes to verify closure.

9. Limitations

The target is a simulated host on loopback, so service banners reflect how the lab services were configured and are not proof of the real software build. OS detection is low confidence for the reason given in 4.4, and UDP scanning covered only the top 100 ports. No exploitation or password guessing was attempted; findings are based on observation and packet analysis only.

10. Conclusion

The audit shows how quickly a few Nmap scans map an attack surface and how a packet capture exposes any protocol that lacks encryption. The most urgent fixes are removing cleartext protocols (Telnet, FTP, HTTP login) and closing or restricting SNMP and MySQL. Following the recommendations and re-scanning would turn this host from one with several trivially exploitable weaknesses into one exposing only encrypted, patched, and necessary services.

Appendix - Evidence files

syn.txt, udp.txt, ver.txt, os.txt, scripts.txt (Nmap normal output); scan.pcap and creds.pcap (packet captures openable in Wireshark).